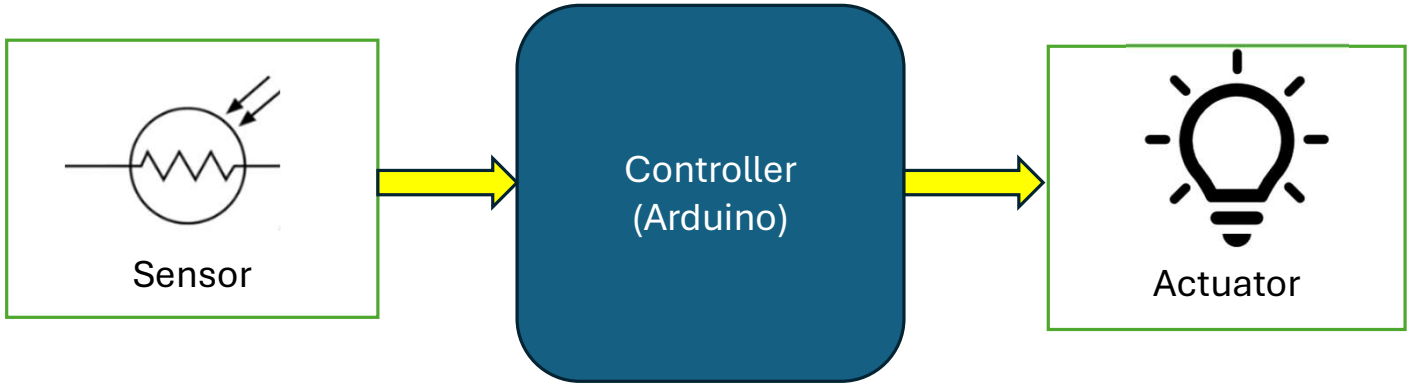
An Introduction to Sensors

The Eyes and Ears of Modern Technology

Sensors

- Introduction to Sensors
- Sensor Classifications, types
- Sensor examples

Automatic Streetlight System



The streetlight automatically turns **ON at night** and **OFF during the day** without human effort.

- **Sensor** → Light Dependent Resistor (LDR) detects ambient light level.
- **Controller** → Simple microcontroller or comparator circuit decides whether it is day or night.
- **Actuator** → Relay + Lamp (turns light ON/OFF).

Sensors

- Sensor is a device that **detects a physical quantity** and converts it into a **measurable signal**.

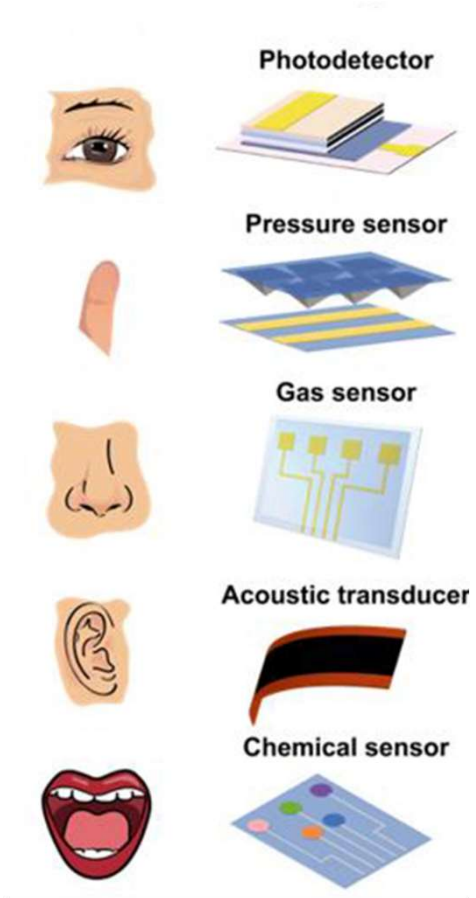


Physical quantity (stimulus):
Temperature, displacement, humidity, force, acceleration…

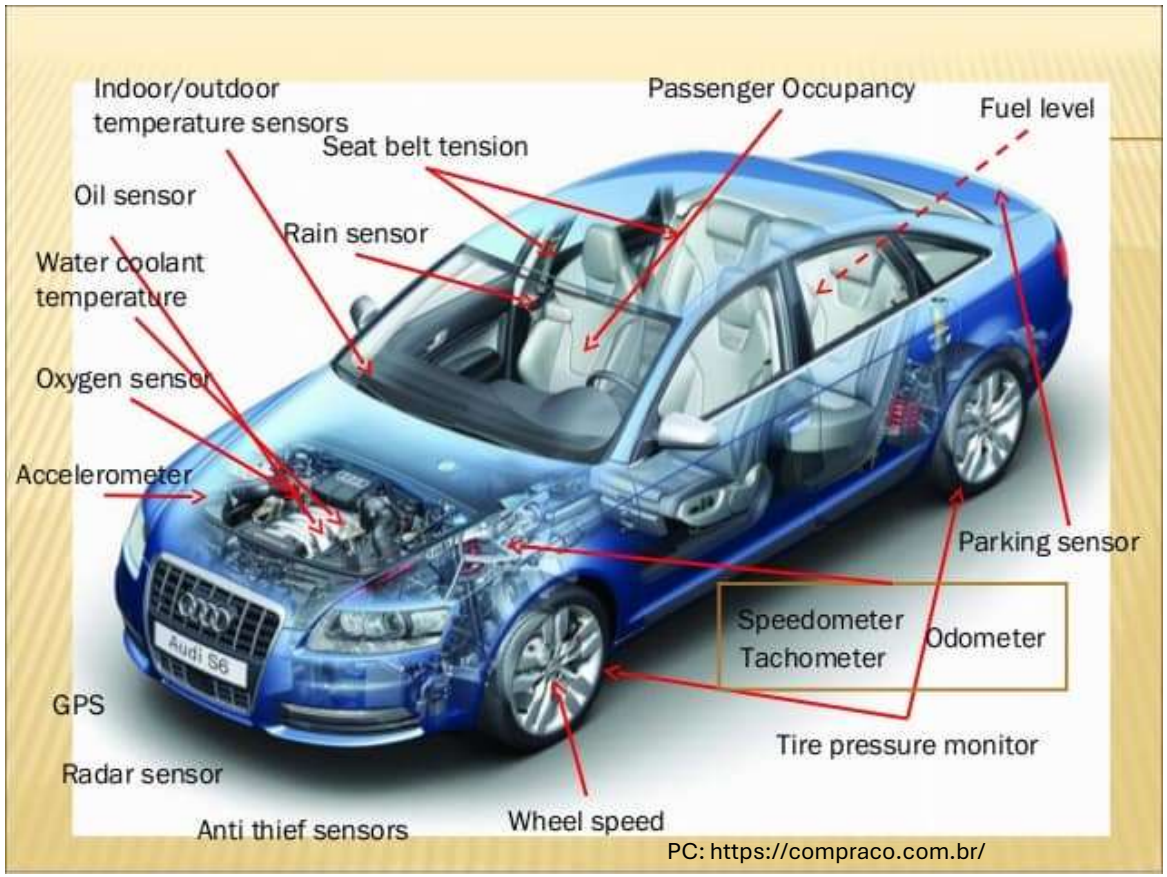
Measurable signal (Output signal):
Electrical (voltage, current, resistance), Mechanical, Optical,…

Human sensing Vs Electronic Sensors

Human Sense	What We Perceive	Equivalent Sensor Examples
👁️ Vision	Light / Images	Light sensors (LDR, camera sensor
👂 Hearing	Sound / Vibrations	Noise detector, Acoustic transducer
👃 Smell	Chemicals / Odors	Gas sensor
👅 Taste	Chemicals / Flavor	Chemical/biosensors
✋ Touch	Pressure / Temperature / Pain	Pressure sensor, temperature sensor, tactile sensor



Sensors in Daily Life Examples – Car Sensors



PC: https://compraco.com.br/

- **Sensors for engine control**
- **Safety- Security Sensors**

Sensors in Daily Life Example – Sensors in Smartphone



Sensor Classifications

- Sensor classification schemes range from very simple to the complex. Depending on the classification purpose, different classification criteria may be selected.

  - ❑Digital or Analog
  - ❑Active or Passive
  - ❑Classification based on the stimulus (Physical Quantity measured)

Sensor Classifications: **Digital Vs Analog**

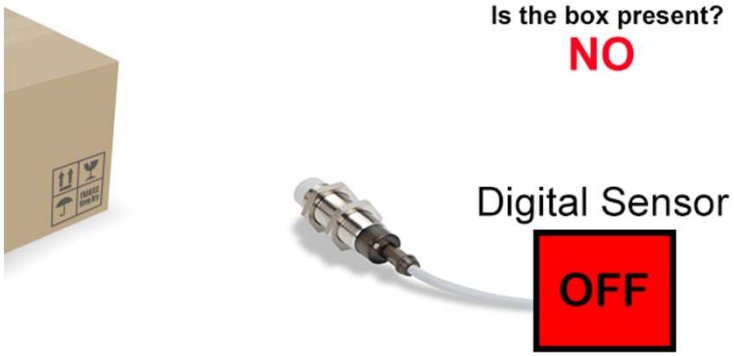
- Sensors are classified as ***analog*** or ***digital*** based on the **type of output signal**

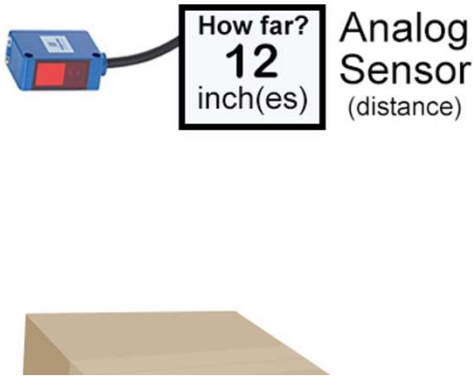
- **Digital:** Output is discrete values

  These sensors supply an **ON or OFF** signal

- **Analog:** Output is continuous values

  Analog sensors take a physical quantity and convert it to electrical signal with typical analog signals ranging between **0-10VDC** or **4-20mA.**

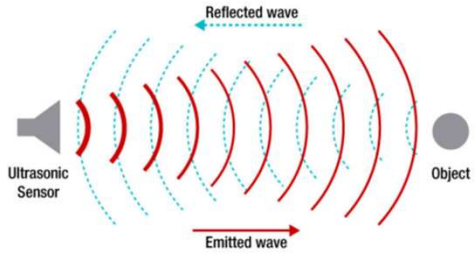




pc: https://library.automationdirect.com

Sensor Classification: **Active and Passive**

- **Active sensors emit energy into the environment to facilitate a measurement.**
- **Active sensors** are characterized by their active role in the measurement process. Think of them as actively "probing" their surroundings.
- This energy could be in the form of radio frequency (RF) waves, acoustic waves, or even light, which then reflects off the object or phenomenon being measured and is picked up by the sensor.
- The **active sensors** require external power source
- Ex: Ultrasonic sensor, LIDAR



- **Passive sensors do not actively emit energy into the environment.**
- They rely on receiving naturally available energy or changes in that energy. This means they measure the energy that is already present and then translate it into a signal.
- Passive sensors do not require any additional power source
- Ex: Thermocouple, photodiode, piezoelectric sensor



Passive Vs. Active Sensor

Passive sensors respond to the physical stimulus without requiring an external power source.

- **Energy Source:** Passive sensors do not require an external power supply to operate. They derive their energy from the physical phenomenon they are measuring.
- **Output:** The output of passive sensors is usually analog in nature, directly proportional to the measured quantity.
- **Applications:** Passive sensors are commonly used in applications where external power sources might not be available or practical, such as in remote or harsh environments.
- Examples: Thermocouples, resistive temperature detectors (RTDs), and strain gauges

Active sensors actively emit a signal or energy and then measure the response to that signal after it interacts with the environment.

- **Energy Source:** Active sensors require an **external power source** to generate the signal they emit.
- **Output:** The **output** of active sensors **can be digital or analog**, depending on the **sensing mechanism and the processing** involved.
- **Applications:** Active sensors are used in situations where the object to be measured is far away, obscured, or where passive measurements might not provide sufficient accuracy. They are commonly used in distance measurement, object detection, and remote sensing applications.
- Examples: Ultrasonic sensors, radar sensors, and infrared sensors.

Sensor Classification – **Stimuli Types**

Note: The list is by no mean comprehensive!

- **Linear/Rotational sensors**
  Measure displacement or angle (e.g., potentiometer, rotary encoder).
- **Force, torque, and pressure sensors**
  Detect applied loads, twists, or fluid/gas pressure (e.g., strain gauge, load cell).
- **Acceleration sensors**
  Sense motion and vibration (e.g., accelerometer in phones, airbags).
- **Flow sensors**
  Measure liquid or gas movement (e.g., water flow meter, airflow sensor).
- **Temperature sensors**
  Detect heat or cold (e.g., thermocouple, thermistor).
- **Proximity sensors**
  Detect presence without contact (e.g., IR sensor, ultrasonic sensor).
- **Light sensors**
  Measure brightness or light intensity (e.g., LDR, photodiode).
- **Smart material sensors**
  Use special materials that respond to stimuli (e.g., piezoelectric vibration sensors, shape-memory alloys).

Sensor Example: Proximity Sensor

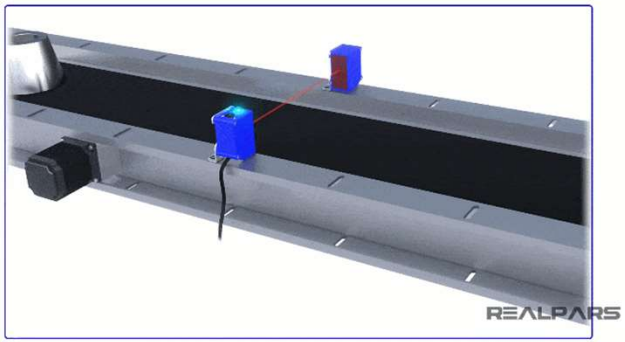
- They are used to sense the **proximity of an object** relative to another object.
- They are **non-contact sensors**
- They usually provide a on or off signal indicating the presence or absence of an object.

Common Types:

- ***Capacitive proximity sensor:*** The proximity of an object changes the gap and affects the capacitance.
- ***Inductive proximity Sensors***: The inductance of the sensor changes when a ferrous object is in its proximity
- ***Photoelectric:*** The proximity of a moving object interrupts the light beam causing the voltage level to change.



Capacitive/Inductive types



Photoelectric sensor

Applications:

- Can be used on conveyor lines for object detection
- Liquid level measurement
- Material handling, food industry,…

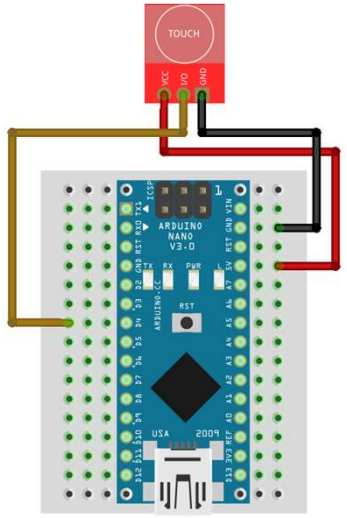
Sensor Example: Tactile (touch) Sensor

A **tactile sensor** is a type of sensor that detects and measures information through **physical contact** with an object or surface.

It detects **touch, force or pressure** on its surface changing the logic state of the circuit. It works in very similar way as to buttons.

Tactile Sensor types examples:

- **Resistive Tactile Sensors:** Pressure or force changes resistance.
- **Capacitive Tactile Sensor**: Pressure changes the distance between plates → capacitance changes.
- **Piezoelectric Tactile Sensors:** Mechanical stress produces a small voltage.



PC: https://projecthub.arduino.cc/

This example demonstrates the use of a touch sensor with *Arduino Nano* .

By default, the signal value is *LOW* and it switches to *HIGH* when the touch sensor is pressed.

Sensors vs. Transducers vs. Actuators

- A **transducer** is a device that **converts energy from one form to another.**

- **All sensors are transducers**, because they convert a physical quantity (like temperature, light, or pressure) into another form (often electrical).
- But **not all transducers are sensors** - some are actuators (they take electrical energy and convert it into motion, sound, etc.)

Sensor → measures physical input
Transducer → converts one form of energy to another
Actuator → produces action/output

Examples of Transducers:

Sensor-type (input transducers):
- **Microphone** → converts sound (acoustic energy) → electrical signal.
- **Thermocouple** → converts temperature → voltage.
- **Photodiode** → converts light → electrical current.

Actuator-type (output transducers):
- **Speaker** → converts electrical signal → sound.
- **Electric motor** → converts electrical energy → mechanical rotation.
- **LED** → converts electrical energy → light.